INTRODUCTION

Chemistry touches every aspect of our lives. The three-basic requirement of our life: food, clothes, shelter are all basically chemical compounds. Infact, life itself is a complicated system of interrelated chemical process. In this unit, we will learn the chemistry involved in the field of medicines, food materials, cleansing agents and polymers.

15.1 Drug

The word drug is derived from the French word "*drogue*" meaning "*dry herb*". A drug is a substance that is used to modify or explore physiological systems or pathological states for the benefit of the recipient. It is used for the purpose of diagnosis, prevention, cure/relief of a disease.The drug which interacts with macromolecular targets such as proteins to produce a therapeutic and useful biological response is called medicine. The specific treatment of a disease using medicine is known as chemotherapy. An ideal drug is the one which is non-toxic, bio-compatible and bio-degradable, and it should not have any side effects. Generally, most of the drug molecules that are used now a days have the above properties at lower concentrations. However, at higher concentrations, they have side effects and become toxic. The medicinal value of a drug is measured in terms of its therapeutic index, which is defined as the ratio between the maximum tolerated dose of a drug (above which it become toxic) and the minimum curative dose (below which the drug is ineffective). Higher the value of therapeutic index, safer is the drug.

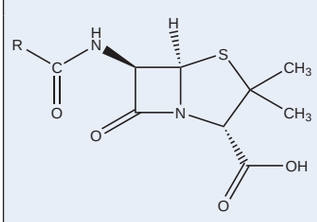
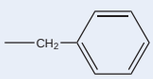
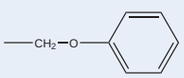
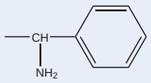
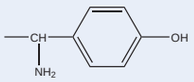
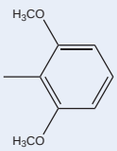
15.1.1 Classification of drugs:

Drugs are classified based on their properties such as chemical structure, pharmacological effect, target system, site of action etc. We will discuss some general classifications here.

Classification based on the chemical structure:

In this classification, drugs with a common chemical skeleton are classified into a single group. For example, ampicillin, amoxicillin, methicillin etc.. all have similar structure and are classified into a single group called penicillin. Similarly, we have other group of drugs such as opiates, steroids, catecholamines etc. Compounds having similar chemical structure are expected to have similar chemical properties. However, their biological actions are not always similar. For example, all drugs belonging to penicillin group have same biological action, while groups such as barbiturates, steroids etc.. have different biological action.

Penicillins

	R group-Drug Name				
	 Penicillin G	 Penicillin V	 Ampicillin	 Amoxicillin	 methicillin

XII U15 Chemistry in Action.indd 273 11-01-2021 17:15:56